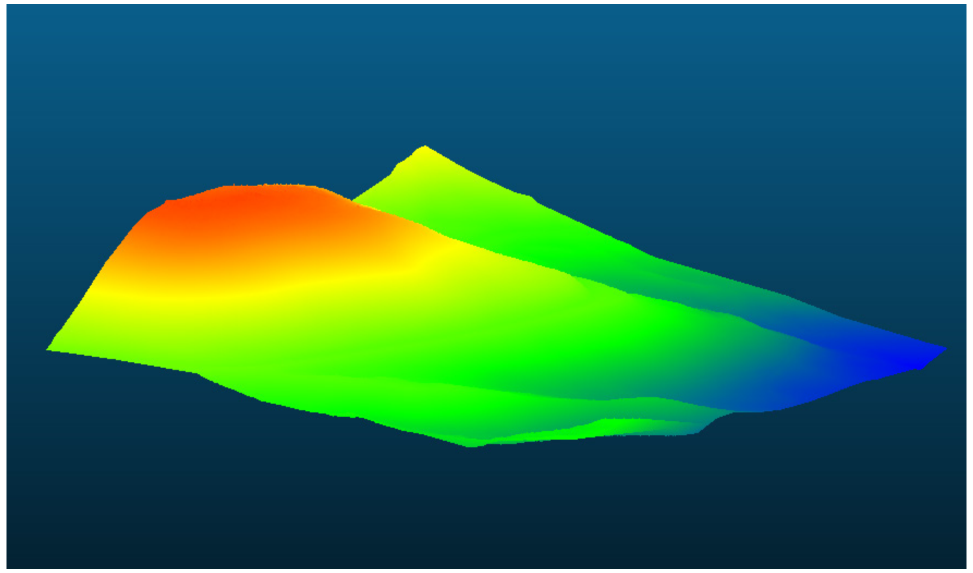
Advanced Computer Applications

ASSIGNMENT 1

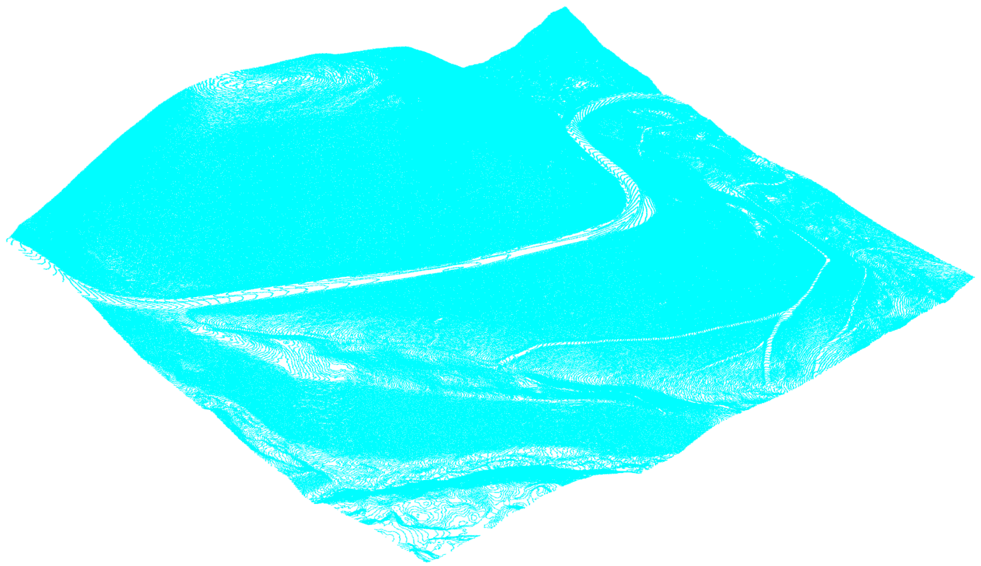
Scanning and Data Processing

Project Description

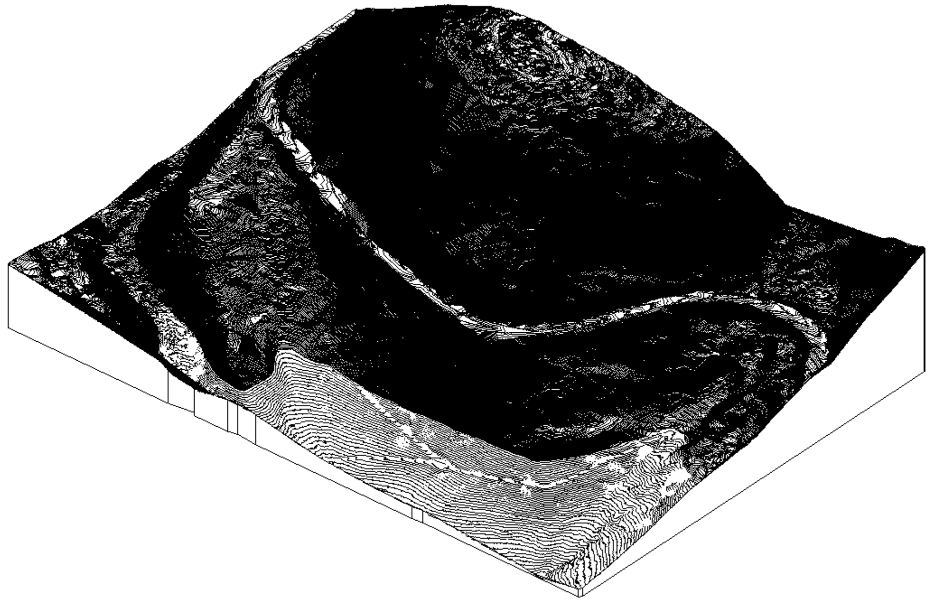
For this assignment, I selected a temporary site since our studio location had not yet been determined. The site is along the road to Big Sky Resort. I obtained LiDAR data from OpenTopography, which includes point data for both the ground surface and off-ground elements such as trees and vegetation. To generate an accurate terrain model, I processed the point cloud in CloudCompare. Using CSF filter, I separated the off-ground points from the ground points, resulting in a clean dataset representing only the site topography. I then generated a mesh from the ground points and exported it as an OBJ file for use in Rhino. Within Rhino, I prepared the geometry to be exported as a DWG file. This file was imported into Revit, where I used the Toposolid from DWG command to create a Revit native topo surface. This workflow works super well for obtaining super detailed topography.



Cleaned Ground-Points via Cloud Compare



Cleaned Ground-Points Mesh (Rhino)

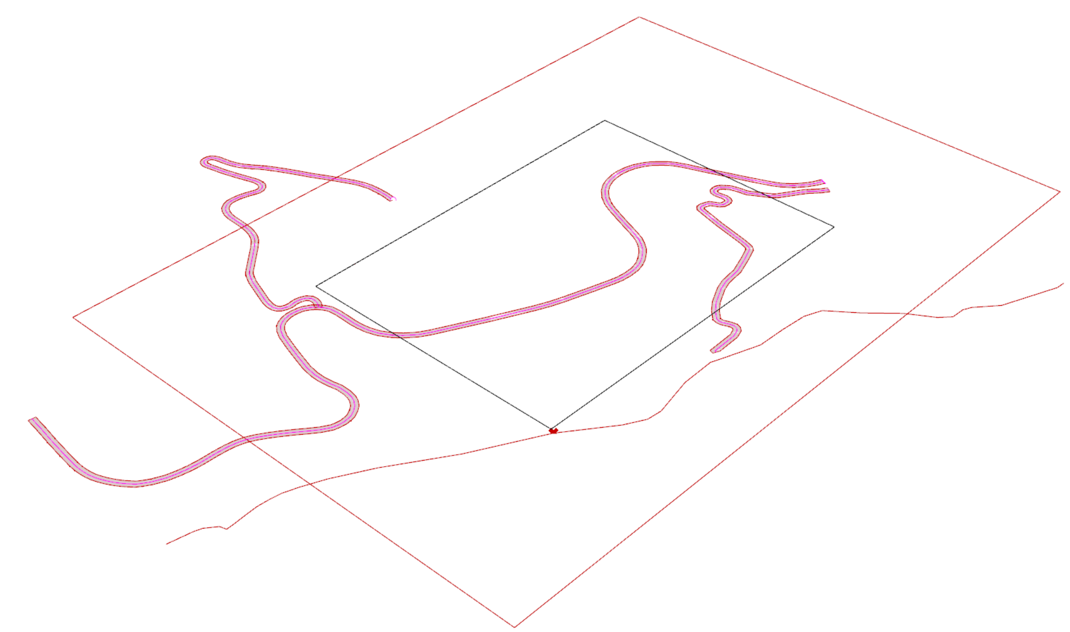


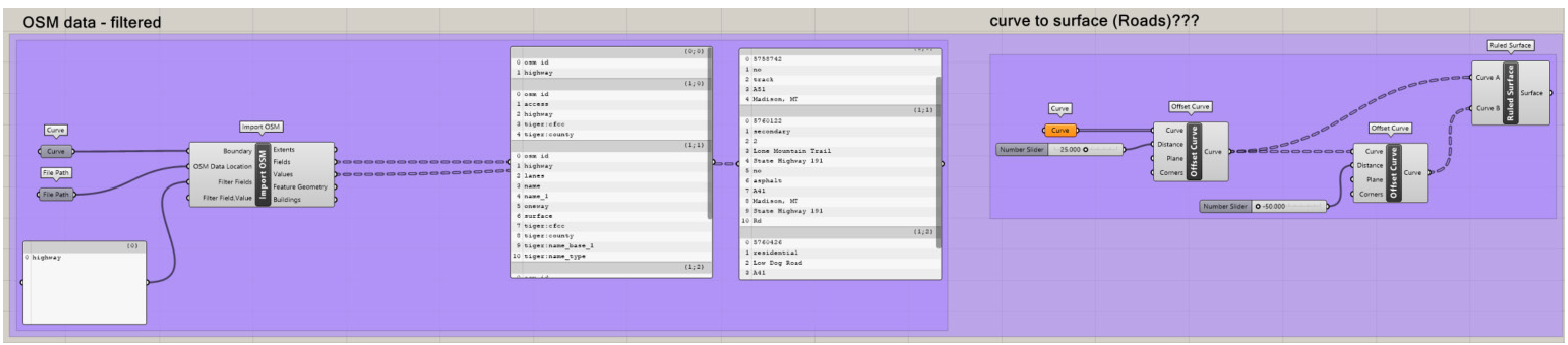
Revit Topo-Solid

OSM Data

Project Description

For this portion of the assignment, I used Grasshopper to import OpenStreetMap (OSM) data. The dataset contains multiple layers of information that can be filtered directly within Grasshopper. We isolated the highway network to focus on primary circulation.
Max and I then explored how to develop this line work into a more detailed model. Using Grasshopper, we parametrically assigned a realistic width to the road geometry, transforming the centerlines into surface representations. The next step would be to project these road surfaces onto the topographic mesh. From there, the geometry could be given thickness and further detailed with elements such as curbs, edge conditions, or other site-specific components.





Rhino Topo to Laser Cut File

Project Description

In developing the final mesh-to-laser-cut workflow, we were given a complete Grasshopper script that converts a mesh into a set of laser-cut paths, automating the file preparation process. While running our model through the script, Max and I encountered an issue where the output geometry was producing open curves rather than closed cutting paths. To address this, we worked through the script to locate where the breaks were occurring. I tested a series of fixes using List Item to isolate problematic segments, Extend Curve to close small gaps, and then Rejoin to rebuild continuous lines. Although the solution was only partially successful, the process helped me better understand how to read the data tree, trace errors through the script, and identify where geometric continuity is lost. This process was valuable in learning how to diagnose issues in the script rather than relying solely on the provided function of it.